

Dividend Pricing Rule Methodology

Equity Derivatives Trading - Intern Project - August 24, 2026

Overview

The Dividend Pricing Rule provides a systematic pricing shift for single-stock dividend risk. It combines three quantitative pillars: a **cancellation penalty** for names with recent defaults, an **implied volatility scale** from consensus prices, and a **sector-level Dividend Risk Premium (DRP)** derived from dividend swap discounts.

Core Formulation

$$\text{Dividend Pricing Rule}_i = \begin{cases} -\sigma_i \times [1 \pm (k_i \times \text{DRP}_{s,t})] & \text{if } C_i = 0 \\ -\frac{1}{n} & \text{if } C_i = 1 \\ -1 & \text{if } C_i > 1 \end{cases}$$

Specifications

1. Flat Cancellation Shift ($C_i > 0$)

- **Target Universe:** Names with a dividend cancellation in the trailing 3-year lookback window that have subsequently resumed payments.
- **Mechanism:** Penalizes recurring cancellation risk by scaling the downward shift based on payout frequency (n) and missed payments (C_i):
 - **Single Cancellation** ($C_i = 1$): Mark shifts downward by one payment period ($-\frac{1}{n}$).
 - **Multiple Cancellations** ($C_i > 1$): Full -100% shift (-1).

2. Implied Cash Dividend Volatility (σ_i)

- **Target Universe:** Active dividend payers with no recent cancellations ($C_i = 0$).
- **Mechanism:** Captures dividend amount uncertainty in dollar terms using a 1-year (252d) rolling standard deviation of daily Totem consensus prices.
- **Tenor Truncation:** Restricted strictly to 1–3Y tenors to eliminate illiquidity noise, sparse observations, and extreme variance typical of 5–7Y marks.

3. Sector-Level Dividend Risk Premium ($\text{DRP}_{s,t}$)

- **Mechanism:** Quantifies the structural discount at which implied dividends (via dividend swaps at $t - 1$) trade relative to realized dividends (at t):

$$\text{DRP}_{s,t} = \frac{\text{Realized}_{s,t} - \text{Implied}_{s,t-1}}{\text{Realized}_{s,t}}$$

- **Premium Adjustment:** Scaled by underlying parameter k_i . The $\pm$ direction adjusts for the desk's position (long vs. short dividends).

Parameter Reference

C_i = Number of dividend cancellations for asset i in the trailing 3-year window

n = Annual dividend payment frequency (e.g., 4 = Quarterly, 2 = Semi-Annual)

σ_i = 1-year rolling standard deviation of daily Totem consensus marks (1–3Y tenors)

$DRP_{s,t}$ = Dividend Risk Premium for sector $s = \text{sector}(i)$ at period t

k_i = DRP scaling factor for asset i